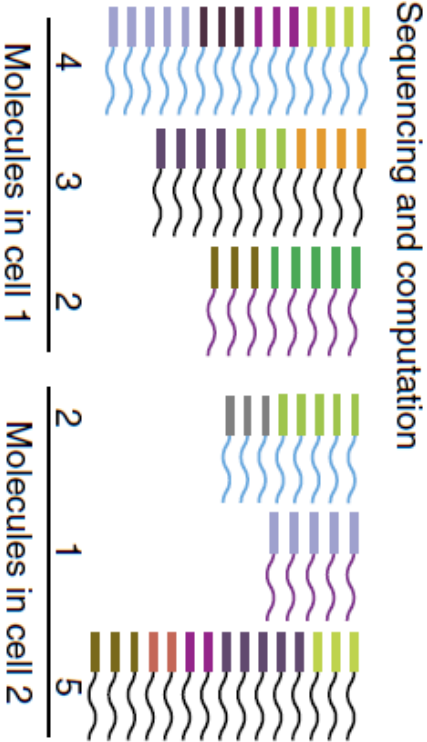
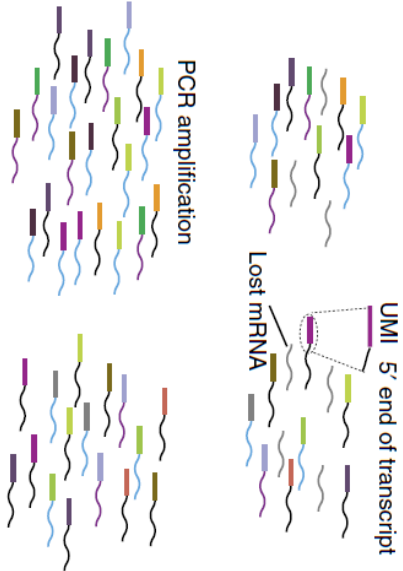
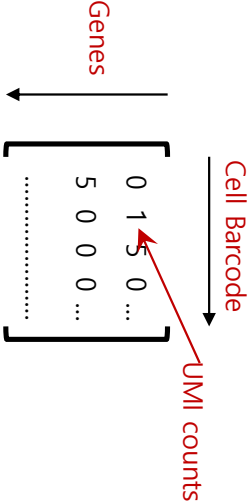


Unique Molecular Identifier (UMI)



PMID 28091601 & 24363023

AGCTTATGCT	Gene A	O
AGCTTATGCT	Gene A	X
TTGATGCTAG	Gene A	O



Keywords

Single cell RNA sequencing

Cellular identity and states at a transcriptome level

Drop-out

Cell barcodes

UMI

Full-length, 5', 3' sequencing

CITE-seq

Spatial transcriptomics

single cell BCR/TCR sequencing

Cell Ranger

Normalization

Scaling, z-transformation

Principal component analysis

Data visualization, PCA, UMAP, tSNE

Cell type annotation, cluster DEG